MACROECONOMICS PRELIM, JULY 2025

ANSWER KEY FOR QUESTIONS 1 AND 2 (ECN 200D)

Question 1

a) A RCE for this economy is a list of:

$$\begin{gathered}\text{pricing functions: } r(\bar{k}), w(\bar{k}) \\ \text{policy function: } k' = g(k, \bar{k}) \\ \text{value function: } V(k, \bar{k}) \\ \text{transition function: } \bar{k}' = G(\bar{k})\end{gathered}$$

such that:

1) $g(k, \bar{k})$ and $V(k, \bar{k})$ solve the agent's recursive problem

$$\begin{aligned} V(k, \bar{k}) &= \max_{c,k'} \{u(c) + \beta V(k', \bar{k}')\}, \\ s.t.\;\; c + k' &= w(\bar{k}) + \left[r(\bar{k}) + 1 - \delta\right] k, \\ \bar{k}' &= G(\bar{k}). \end{aligned}$$

2) Prices are competitively determined:

$$\begin{aligned} r(\bar{k}) &= F_1(\bar{k}, \bar{k}) = aA\bar{k}^{a+\gamma-1}, \\ w(\bar{k}) &= F_2(\bar{k}, \bar{k}) = (1-a)A\bar{k}^{a+\gamma}. \end{aligned}$$

3) Aggregate consistency:

$$G(\bar{k}) = g(\bar{k}, \bar{k}).$$

Solving the agent's recursive problem (in any way you find convenient) will lead to the following Euler equation:

$$u'(c) = \beta \left[r(\bar{k}') + 1 - \delta\right] u'(c').$$

Typically, at this point we can replace the consumption expressions above from the budget constraint and describe the RCE more sharply (as we did in class). However, since the question explicitly asks you to describe only the steady state capital, we can focus on steady states right away, and re-write the last Euler equation as

$$1 = \beta \left[r(\bar{k}) + 1 - \delta\right].$$

We can now replace the rental rate of capital from the definition of the RCE above, and after a little algebra we will arrive at the desired result. More specifically, we will find that the steady state level of capital in this economy with production externalities will be given by

$$\bar{k} = \left[\frac{1-\beta(1-\delta)}{a\beta A}\right]^{\frac{1}{a+\gamma-1}}.$$

b) We now move on to the problem of the social planner who, as we described in the question, internalizes the production externality. The recursive problem for the social planner is as follows:

$$\begin{aligned} V(\bar{k}) &= \max_{c,\bar{k}'}\{u(c) + \beta V(\bar{k}')\}, \\ s.t.\ c + k' &= A\bar{k}^{a+\gamma} + (1-\delta)k. \end{aligned}$$

Solving the social planner's recursive problem (in any way you find convenient) will lead to the following Euler equation:

$$u'(c) = \beta\left[(a+\gamma)A\bar{k}^{a+\gamma-1} + 1 - \delta\right]u'(c').$$

Once again we will focus right away on steady states, and this will allow us (after some algebra) to immediately calculate the steady state level of capital that the hypothetical social planner would have chosen:

$$k^S = \left[\frac{1-\beta(1-\delta)}{(a+\gamma)\beta A}\right]^{\frac{1}{a+\gamma-1}}.$$

Our last task in this part is to compare this level of capital with the one attained in the decentralized economy (of part (a)), where agents do not internalize the production externality. Of course, our intuition strongly suggests that the social planner, who internalizes the externality, would choose a higher level of capital. Our intuition will be confirmed if and only if

$$\bar{k} < k^S \Leftrightarrow \left[\frac{1-\beta(1-\delta)}{a\beta A}\right]^{\frac{1}{a+\gamma-1}} < \left[\frac{1-\beta(1-\delta)}{(a+\gamma)\beta A}\right]^{\frac{1}{a+\gamma-1}}.$$

Notice that one of the maintained assumptions of the model is that $\gamma < 1 - a$, implying that $a + \gamma - 1 < 0$, which, in turn, means that the powers to which these expressions are risen are negative, and the result will be confirmed if and only if

$$\frac{1-\beta(1-\delta)}{a\beta A} > \frac{1-\beta(1-\delta)}{(a+\gamma)\beta A} \Leftrightarrow a + \gamma > a,$$

which, of course, is true. Thus, the math confirms our strong conjecture that the social planner's steady state capital should be higher. In fact, it also tells us that it will be higher as long as $\gamma > 0$, since $\gamma = 0$ would simply capture the case where the externality completely disappears.

c) We now move to the (decentralized) economy where the government imposes a lump-sum tax, and uses the tax revenue to subsidize investment (hoping that in this way they will be able to incentivize agents to make the "correct" investment decisions). In this new environment, a RCE is a list of:

$$\begin{gathered}\text{pricing functions: } r(\bar{k}), w(\bar{k})\\ \text{policy function: } k' = g(k,\bar{k})\\ \text{value function: } V(k,\bar{k})\\ \text{transition function: } \bar{k}' = G(\bar{k})\\ \text{taxation function: } T(\bar{k})\end{gathered}$$

such that:

1) $g(k,\bar{k})$ and $V(k,\bar{k})$ solve the agent's recursive problem:

$$\begin{aligned} V(k,\bar{k}) &= \max_{c,k'}\{u(c) + \beta V(k',\bar{k}')\},\\ s.t.\ \ c + (1-\tau)\left[k' - (1-\delta)k\right] &= w(\bar{k}) + r(\bar{k})k - T(\bar{k}),\\ \bar{k}' &= G(\bar{k}).\end{aligned}$$

2) Prices are competitively determined:

$$\begin{aligned} r(\bar{k}) &= F_1(\bar{k},\bar{k}) = aA\bar{k}^{a+\gamma-1},\\ w(\bar{k}) &= F_2(\bar{k},\bar{k}) = (1-a)A\bar{k}^{a+\gamma}.\end{aligned}$$

3) Aggregate consistency:

$$G(\bar{k}) = g(\bar{k},\bar{k}).$$

4) The government's budget constraint is satisfied in every period:

$$T(\bar{k}) = \tau\left[\bar{k}' - (1-\delta)\bar{k}\right].$$

d) In the last part, we need to calculate the subsidy rate that would incentivize agents to fully internalize the production externality, thus, equating the steady state level of capital in the decentralized economy with that of the social planner (described in part (b)). First, we need to solve the new recursive problem of the typical agent, described earlier in part (1) of the definition of the RCE. Solving that problem (in any way you find convenient) will lead to the following Euler equation:

$$(1-\tau)u'(c) = \beta\left[r(\bar{k}') + (1-\tau)(1-\delta)\right]u'(c').$$

Once again, here there is no reason to develop this Euler equation any further in order to characterize the RCE more sharply. We will simply focus on steady states right away, and re-write the last Euler equation as

$$1-\tau=\beta\left[aA\bar{k}^{a+\gamma-1}+(1-\tau)(1-\delta)\right].$$

We can now solve for the steady state level of capital in this economy with production externalities *and* government intervention, and, eventually, find that it is given by:

$$\bar{k}=\left[\frac{1-\tau-\beta(1-\tau)(1-\delta)}{a\beta A}\right]^{\frac{1}{a+\gamma-1}}.$$

The level of government subsidies that would equalize this level of capital with the one chosen by the social planner, is the τ that equalizes the above expression with the expression for k^S provided in part (b). That is, we are looking for the τ that solves:

$$\left[\frac{1-\tau-\beta(1-\tau)(1-\delta)}{a\beta A}\right]^{\frac{1}{a+\gamma-1}}=\left[\frac{1-\beta(1-\delta)}{(a+\gamma)\beta A}\right]^{\frac{1}{a+\gamma-1}}.$$

After some easy algebra, we finally obtain the unique value of τ that achieves the desired result, and it is given by:[1]

$$\tau=\frac{\gamma}{a+\gamma}.$$

[1] In questions like this, it is always good to perform a logical consistency test. Logic and intuition suggest that if γ was equal to 0, then the government should have no reason to intervene. Indeed, we can see in the formula above that if γ is equal to zero, the same should be true for τ. This type of test should make you very confident that the result you found is correct. (Or, if the result makes no sense, it could help you identify that you have made a mistake and try again!)

Question 2

a) The typical buyer's enters the CM with some money, m, and some debt, say d. Her value function is given by

$$W(m,d) = \max_{X,H,\hat{m}} \Big\{ X - H + \beta V(\hat{m}) \Big\},$$
$$s.t. \;\; X + \phi\hat{m} = \phi m + H - d + T,$$

where T is the lump-sum monetary transfer to the buyer by the monetary authority. Using the standard procedure described in class, we will find that

$$W(m,d) = Y + \phi m - d,$$

where Y just summarizes a number of terms that are not related to the state variables.

As we know from class, the sellers' VF are also linear, and it ends up being equal to:

$$W^S(m,d) = \phi m + d.$$

b) Bargaining in a type-N meeting is identical to the one we saw in class. The buyer wishes to purchase the first-best amount q^*, but she may be constrained by her cash holdings. The quantity exchanged in this meeting is given by $q = \min\{\tilde{q}(m), q^*\}$, where

$$\tilde{q}(m) \equiv \{q : z(q) = \phi m\},$$

and, as usual, $z(q)$ has been defined as

$$z(q) \equiv \theta q + (1-\theta)u(q).$$

The amount of dollars that change hands, call it x, solves $x(m) = \min\{z(q^*)/\phi, m\}$.

In a type-C meeting, the buyer has unlimited unsecured credit. Thus, it would be irrational to not trade the first-best quantity q^*. In return, the buyer will make a promise to repay $z(q^*)$ units of the numeraire in the forthcoming CM round of trade.

c) As is typically the case in these types of questions, you can derive the objective function in any way you like, and this includes a good guess. Here the objective function is simply given by:

$$J(m') = (-\phi + \beta\phi')m' + \beta(1-\Lambda)[u(q(m')) - \phi' x(m')].$$

This looks almost identical to the J function we derived in class, except here the liquidity premium term is multiplied by $1-\Lambda$ because this is the probability of meeting an N-type seller (and money is useful only in that scenario). Now, we also know that in equilibrium $\phi \geq \beta\phi'$, which means that the buyer will never bring more money that what is necessary to purchase q^*. Thus, it is safe to focus on the "binding part" of the bargaining protocol, and re-write our objective as:

$$J(m') = (-\phi + \beta\phi')m' + \beta(1-\Lambda)[u(\tilde{q}(m')) - \phi' m'].$$

d) The next step is to obtain the FOC and evaluate it in equilibrium. We have:

$$\phi = \beta\phi' + \beta(1-\Lambda)\left[u'(\tilde{q})\frac{d\tilde{q}}{dm'} - \phi'\right].$$

To obtain an expression for how $\tilde{q}$ changes as the buyer brings an additional dollar, apply the IFT on the definition of $\tilde{q}$, as we did in class. This will allow us to write:

$$\phi = \beta\phi'\left\{1 + (1-\Lambda)\left[\frac{u'(\tilde{q}(m'))}{z'(\tilde{q}(m'))} - 1\right]\right\}.$$

In the steady state equilibrium, we have $z = \phi M = \phi' M'$. Moreover, using the Fisher equation, we can replace the term $(1+\mu-\beta)/\beta$ with the nominal interest rate on an illiquid bond, i. Hence, the earlier FOC will become:

$$i = (1-\Lambda)\left[\frac{u'(q)}{z'(q)} - 1\right]. \tag{1}$$

A steady state equilibrium in this model is a triplet (q, z, q_C), such that q solves the equilibrium condition (1), the real balances satisfy $z = z(q)$, and $q_C = q^*$ always, since, as we already discussed, it would be irrational to not purchase the first best in a meeting with a C-type seller.

e) Clearly, a seller who decides to pay the κ cost enjoys a net benefit of

$$S_C = z(q^*) - q^* - \kappa.$$

As is standard, $z(q^*)$ is the seller's payment derived through bargaining, and here the seller's cost is linear. Of course, a C-type seller always produces the first best. Clearly, this term does not depend on what other sellers do (i.e., it does not depend on Λ).

f) A seller who chooses to not pay κ enjoys a steady state benefit of

$$S_N = z(q(\Lambda)) - q(\Lambda).$$

This seller is better off because she skipped paying the information cost, but she is worse off because she is now an N-type, and how much she will sell depends on Λ, as equation (1) makes perfectly clear. The economics is pretty obvious: if I am an N-type my sales depend on how much money buyers bring, which, in turn, depends on how widespread credit is in this economy (i.e., the value of Λ). Intuition suggests that as Λ goes up, buyers should bring less money. And as $\Lambda \to 1$, buyers should not carry any money, since carrying money is costly, and it is extremely unlikely they would ever use it. Formally:

$$\frac{S_N}{d\Lambda} = z'(q)q'(\Lambda) - q'(\Lambda) = q'(\Lambda)[z'(q) - 1] = q'(\Lambda)[u'(q) - 1](1 - \theta).$$

The sign of the last expression coincides with the sign of $q'(\Lambda)$ because the other two terms are positive. Of course, equation (1) (the money demand function) immediately implies that the sign of $q'(\Lambda)$ is negative. This confirms our earlier conjecture.

g) If $\Lambda = 1$ buyers bring no money and a seller who decides to stay an N-type makes zero profit. Assuming that κ is not crazy high (specifically, assuming that $\kappa < z(q^*) - q^*$), the smart seller will decide to become a C-type, or, in math, $\lambda^*(1) = 1$.

h) As we have already established, when $\Lambda = 0$ the representative seller's profit from not paying the cost is maximized (because buyers' money demand is maximized). To make the math easier, I also hinted that you should focus on the case $i \approx 0$. When both Λ, i are 0, equation (1) implies that $q = q^*$. In this case,

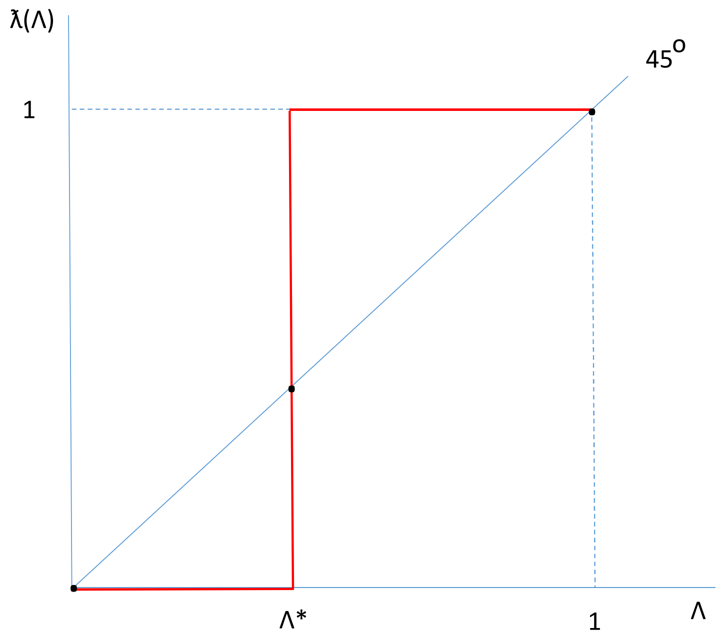
$$S_N = z(q^*) - q^* > z(q^*) - q^* - \kappa = S_C,$$

and, of course, the seller would not pay the cost, or, in math $\lambda^*(0) = 0$.

i) Putting all this information together, we can summarize the representative seller's optimal behavior (as a function of her belief about the behavior of other sellers) by the red segments in the figure below. This figure, which is drawn assuming that $i \approx 0$, should remind you a lot of the figure we drew when we analyzed the Kiyotaki and Wright model. If the seller believes that other sellers are paying κ with high probability, her best response is to also pay that cost. On the other hand, if the seller believes that other sellers are unlikely to pay κ, her best response is to also skip the cost. In between, there is a $\Lambda^* \in (0, 1)$ such that the representative seller is indifferent between paying and not paying. Since an equilibrium is a point where the representative seller's best response correspondence intersects with the 45 degree line, we have three equlibria:

- The full credit equilibrium with $\Lambda = 1$, where all sellers become C-types.
- The no-credit equilibrium with $\Lambda = 0$, where all sellers stay N-types.
- The mixed equilibrium, where a fraction $\Lambda^* \in (0, 1)$ of sellers pay κ and become C-types.

Everything we said so far holds for $i \approx 0$. As i increases, the cost of carrying money also increases and the form of equilibrium may change. First, we know from class that with Kalai bargaining if $i \geq \theta/(1 - \theta)$, no monetary equilibrium will exist. We conclude that there must exist a critical value for the Fisher rate, $\iota \in (0, \frac{\theta}{1-\theta})$, such that:



- If $i \in (0, \iota)$, the there are three equilibria and they are exactly as described above.
- If $i > \iota$, carrying money is too costly, and we have $S_C > S_N$ for all Λ. This implies that the representative seller's best response is $\lambda^*(\Lambda) = 1$, for all Λ, which, in turn, eliminates the monetary equilibrium and implies that the unique equilibrium is the one where $\Lambda = 1$ and all sellers become C-types.